

**Amritraj Lamba**  
lamba.amritraj@gmail.com | +91 8860470606

## **EDUCATION**

**Heritage International Xperiential Learning, Gurugram.**

**Relevant Coursework:** HL Math Analysis and Approaches, HL Chemistry, HL Physics, SL Economics, SL French B, SL English A Language and Literature

**IGCSE (Grade 10, Academic Year 2023-24) :** 10/10 A\*, School Topper in Additional Mathematics, School Topper in Chemistry

**SAT-** 1540

**AP-** Calculus BC: 5/5

**IELTS** - Band 8

## **LEADERSHIP**

**Founder, Sewage Suraksha** **April**  
**2024- Present**

- Developed a sewage lid with multi-gas sensors; it is a low-cost retrofit ensuring safety of municipal workers.
- The lid detects harmful gases, including H<sub>2</sub>S, CH<sub>4</sub>, NH<sub>3</sub>, & CO, and alerts authorities to hazardous gas levels. It has been tested on 17 sewers and has been cleared by the National Disaster Management Authority for adoption.
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**Founder, Flood Shield** **July**  
**2024-Present**

- Flood detection IOT for real-time cloud mapping and emergency coordination; Special Praise for "practicality" at the Princetonian. Adopted by the National Disaster Management Authority.
- Designed and piloted an affordable early warning system for floods that alerts authorities to areas where flooding occurs; adopted by 1000+ houses. Commended by the Haryana Government.
- Also received the Crest Gold Award for this invention

**Founder and Leader—Ingenix School STEM Society** **October 2024-**  
**October 2025**

- Revived and expanded society from 20 to 50+ members. Mentored 2 teams for the Cambridge Science Competition & F1 in Schools, both qualified internationals.

**Leader - "FUNCTION", After School Robotics Club** **October**  
**2024-October 2025**

- Founded a 20+ member club teaching CAD, programming, and mechanical design outside of competition teams
- Introduced collaborative engineering workshops (bridge/tower builds, quizzes) to strengthen teamwork and experimentation
- Built a hands-on learning culture that became an entry pathway for beginners; several members progressed to competitive robotics

## RESEARCH

Automation & Robot Adoption in Delhi-NCR Factories, Author    January 2025-  
September 2025

- Analysed five factories from fully manual (e.g., Metaval Engineering) to highly automated (e.g., Kadence Automation) to assess efficiency, training needs, and scalability
- Found cobot-assisted systems improved efficiency by up to 20% while reducing fatigue and errors; highlighted the need for retraining, ergonomic design, and phased adoption
- Developed a roadmap for SME modernisation without worker displacement; validated through Delhi-NCR field data, mentorship from Stanford's Dr. Arvind Karunakaran, and publication in Lex Localis

Coral Reefs as Natural Barriers to Coastal Erosion in India, Author    March 2024-August 2025

- Used satellite imagery and GIS to show shorelines protected by healthy reefs had less than half the erosion of areas with degraded reefs.
- Provided evidence to integrate reef restoration into climate adaptation strategies for cities like Mumbai and Chennai, complementing engineered defenses.
- Recognition included publication in the International Journal of Environmental Sciences and mentorship from Prof. Umang Soni, with validation in competitive research platforms.

## SUMMER PROGRAMS

**University of Pennsylvania, Wharton Global Youth Program** **June 2025**  
*Product Design Academy*

- Collaborated with Professor Taylor Caputo and international peers under the Wharton faculty, learning design thinking, prototyping and TAM/SAM/SOM analysis.
- Learned to apply structured market-sizing frameworks to test creative engineering ideas against commercial viability. Prototypes developed during the program reflected this rigorous approach, and faculty feedback validated the professional grounding in product innovation.

**LaunchX, Online Bootcamp** **June 2025**

- Selected for LaunchX; collaborated with peers to ideate, prototype, and pitch ventures, gaining hands-on experience in market validation, business modelling, and product design.

## INTERNSHIPS

**Summer Intern, Vardan Envirolabs, Delhi** **December**  
**2024- February2025**

- Identified available and capable employees as part of the talent search module team, designed to assist talent recruitment in companies.
- Conducted market research for the firm, tapping into new domains such as the legal sector, to streamline services and the employment data collection process.

## **ACADEMIC HONORS & AWARDS**

**AMC 12B (2024):** Scored 132/150

- o School Rank 1, Country Rank 28, Qualified for AIME, Certificate of Distinction, Honour Roll (Top 2.5%)

**World International Mathematics Olympiad (WIMO) Final 2024:** Bronze Award  
(Kuala Lumpur, Malaysia)

**CEMC (University of Waterloo):**

- **Pascal Contest (9th Grade, 2023):** 2nd Place from India region, World rank 58, Certificate of Distinction, Student Honour Roll
- **Fryer Contest (9th Grade, 2023):** 2nd Place from India region, Certificate of Distinction
- **Cayley Contest (10th Grade, 2024):** Certificate of Distinction, Student Honour Roll
- **Galios contest (10th Grade, 2024):** Certificate of Distinction
- **Fermat Contest (11th Grade, 2025):** 9th Place from India region, Certificate of Distinction, Student Honour Roll
- **Hypatia Contest (11th Grade, 2025):** Certificate of Distinction
- **Avogadro Chemistry Contest (11th Grade, 2025):** Certificate of Distinction
- **Sir Issac Newton Physics Contest (11th Grade, 2025):** Certificate of Distinction

**First Tech Challenge (FTC) 2024:**

- Role: CAD and Design Head of Team
- Achievements: Qualified for Asian Pacific Open Championship (APOC), Sydney, Australia. Won Inspire Award (2nd Place) and Think Award (2024)

**SIMOC (Singapore International Mathematical Olympiad Challenge) 2024:**

- Maths Olympiad Contest: Silver Award
- Maths Warrior Challenge: Silver Award
- Mind Sports Challenge: Bronze Award

**IOQM (Indian Olympiad Qualifier in Mathematics) 2024–25:** Certificate of Merit

**SASMO (Singapore and Asian School Mathematics Olympiad) 2024:** Gold Award  
(Grade 10)

**AMO (American Mathematics Olympiad):**

- Gold Award (2023, Grade 10)
- Gold Award (2022, Grade 9)

**TIMO (Thailand International Mathematical Olympiad) Final Round 2022–23:**  
Gold Award (Secondary 3 Group)

**HKIMO (Hong Kong International Mathematical Olympiad) Heat Round 2024:**  
Gold Award (Senior Secondary Group)

**Stanford Math Tournament (SMT) 2023:** Honourable Mention in General Test

**Australian Mathematics Competition 2024:** Certificate of Distinction (Senior Division, Grade 11)

**Euler's Circle ( June- July 2024):** One of 30 members worldwide. Capstone on linear algebra and projective geometry; improved modeling intuition later applied to CFD and robot mechanisms. Attended on 50% scholarship

#### **F1 in Schools 2024**

- Role: Design Engineer
- Ranked 15th in Nationals
- Regional Digital Marketing Award

**Future Intelligence Science Olympiad Physics, International round (2024):** Gold Medal

**International Maths Kangaroo Competition (2023):** Silver Medal

**International Junior Honor Society (IJHS), Singapore (2023/2024):** Certificate of Membership

### **SKILLS**

**Technical Skills/Tools:** Python, 3D Modelling (CAD), Arduino, Robotics

**Sports:** Karate Shotokan (Black Belt)

**Music:** Guitar (Trinity Rock & Pop Grade 5)

**Languages:** English, French, Hindi, Punjabi

### **Member**

**New York Academy of Sciences- The Junior Academy member**

Selected member; engaged for a year with the Academy's STEM community. Part of the Solar Power team under Energy Infrastructure.

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